

MUKUL

AI Engineer | RAG & LLM Systems

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PROFESSIONAL SUMMARY

AI Engineer with 1+ year of experience at Tata Consultancy Services building and evaluating LLM-powered retrieval systems. Experienced in developing NLP retrieval pipelines, testing model-assisted outputs for relevance, groundedness, and failure handling, and building full-stack RAG applications end to end. BCA graduate comfortable working across the stack, from data pipelines to production APIs.

WORK EXPERIENCE

Assistant System Engineer, AI/ML Team — Tata Consultancy Services (TCS)

Jun 2025 – Present

- Developed AI-powered support workflows using Python, Flask REST APIs, and React, incorporating a Retrieval-Augmented Generation (RAG) search process over historical incident logs (CSV records covering incident creation, resolver, and resolution details) to help engineers quickly locate similar past incidents.
- Built NLP retrieval pipelines using TF-IDF and BM25 across 2,000+ historical defect records, including text preprocessing and relevance testing with incident queries.
- Fine-tuned and customized existing small language models (SLMs) to power domain-specific Q&A for client platforms (e.g., banking), enabling the model to answer procedural queries with clear, step-by-step responses.
- Reviewed retrieval and model-assisted responses for relevance, factual grounding, and consistency; investigated weak results and refined search or prompting logic.

SELECTED PROJECTS

ResolveIQ — AI Incident Resolution Platform (RAG) — Personal Project

Python, FastAPI, React, BM25, FAISS, Gemini, Docker, GitHub Actions

- Built a full-stack RAG platform combining BM25 and FAISS retrieval with Reciprocal Rank Fusion and optional cross-encoder reranking.
- Added confidence-gated answer generation with Gemini so the system abstains instead of guessing when evidence is weak, with grounded citations on every answer.
- Implemented role-based access, query history, an offline retrieval evaluation harness, and CI/CD with GitHub Actions and Docker.

QueryForge — Natural Language to SQL Platform (SLM) — Personal Project

In Progress

Python, Fine-tuned SLM, SQL

- Building a natural-language-to-SQL platform powered by a fine-tuned small language model that translates plain-English questions into SQL queries.
- Focused on safe, validated query generation to guard against unsafe or destructive SQL execution.

SKILLS

AI Evaluation & NLP: LLM response review, relevance and groundedness checks, prompt testing, citation validation, information retrieval, text preprocessing

Generative AI: Retrieval-Augmented Generation (RAG), LLMs, prompt engineering, embeddings, FAISS, BM25, Gemini

Programming & Tools: Python, SQL, Java, Git, GitHub, Docker, GitHub Actions, Jupyter Notebook, VS Code

Web & Data: FastAPI, Flask, REST APIs, React, MySQL, XML/Excel automation

EDUCATION

Bachelor of Computer Applications (BCA) — Uttaranchal University, Dehradun

2021 – 2024

CGPA: 8.3 / 10